

An 8.91 nW, 1.74 ppm/°C Subthreshold CMOS-Only Voltage Reference for Temperature Sensors

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Abstract—This paper presents a low-power CMOS voltage reference (VR) based on a two-stage stacked diode-connected MOS transistor (SDMT) structure, implemented entirely with MOS transistors. The VR is designed using a standard 0.18 μm CMOS process and occupies an active area of 0.029 mm^2 , with all transistors operating in the subthreshold region. Post-layout simulations show that the reference voltage reaches 365 mV at 27 °C, while the power consumption is only 8.91 nW. The VR exhibits a temperature coefficient of 1.74 ppm/°C across a wide temperature range from −40 °C to 125 °C. In addition, it demonstrates a line sensitivity (LS) of 0.25 %/V over a supply voltage range of 0.7 V to 1.8 V, confirming its excellent stability and ultra-low-power performance for temperature sensor applications.

Keywords—Voltage reference, Subthreshold, Temperature coefficient, Temperature sensor.

I. INTRODUCTION

Voltage reference circuits play a critical role in electronic systems, including analog-to-digital converters, digital-to-analog converters, low-dropout regulators, operational amplifiers, and temperature sensors, by providing stable reference voltages that are resilient to process, voltage, and temperature (PVT) variations. With the rapid advancement of temperature sensor systems, there is an increasing demand for VR circuits capable of stable operation under ultra-low-power conditions. This demand arises because power consumption directly affects system energy efficiency, operational lifetime, and long-term autonomous operation, while voltage stability critically determines the measurement accuracy of temperature sensors [1-4]. Therefore, the development of ultra-low-power and highly stable VRs is essential to enable energy-efficient, reliable, and high-precision temperature sensing in modern integrated systems.

With the increasing demand for temperature sensors, researchers have developed various VR structures that offer excellent temperature stability, sub-1 V supply operation, and sub-microwatt power consumption, addressing the requirements for low-power and high-precision applications [5-8]. Seok et al. proposed a two-transistor VR circuit that achieves stable output without amplifiers, saturating devices, or resistors. The circuit operates at a minimum supply voltage of 0.5 V, consumes only picowatts of power, and occupies an active area of 0.009 mm^2 , while maintaining favorable temperature coefficient (TC), LS, and power-supply rejection ratio (PSRR) [9]. Oliveira et al. further introduced a three-transistor VR, generating the reference voltage from the threshold voltage difference of two cascaded intrinsic NMOS transistors. Utilizing intrinsic MOS devices with extremely low threshold voltages enables a negative gate-source voltage, achieving ultra-low-power operation. Compared to conventional subthreshold designs,

this approach can deliver output voltages below 100 mV and operate reliably at supply voltages as low as 0.12 V, offering significant advantages in power consumption, operating voltage, LS, and area efficiency [10]. Shao et al. presented an SDMT VR circuit with start-up time enhancement. By combining self-biasing with capacitive coupling, the design reduces power consumption and improves PSRR. Measurement results show a power consumption of 1.8 nW and a PSRR of −74 dB at 100 Hz, effectively suppressing supply noise [11]. Agarwal et al. proposed an ultra-low-power VR occupying 0.032 mm^2 . The design leverages the dependence of MOSFET threshold voltage on substrate-to-source voltage to construct a complementary to absolute temperature (CTAT) voltage generator that compensates the proportional to absolute temperature (PTAT) component, achieving a trade-off between low power and high stability [12]. Building on these advances, this work proposes a two-stage SDMT-based subthreshold CMOS-only VR. It features a simple structure, compact area, and ultra-low power consumption, providing an effective solution for reference voltage generation in low-power integrated systems [13-20]. The rest of this paper is organized as follows: Section II discusses the circuit principles and implementation; Section III presents the simulation results; Finally, Section IV provides concluding remarks.

II. DESIGN OF THE PROPOSED VOLTAGE REFERENCE CIRCUIT

The proposed subthreshold CMOS-only voltage reference circuit is shown in Fig. 1 and comprises a start-up circuit, a reference current circuit, and the VR core. Transistors M_{14} and M_{16} are 3.3 V standard NMOS thick-oxide devices with high threshold voltages, while M_9 – M_{12} are 1.8 V native NMOS devices with zero threshold voltage. The remaining transistors are 1.8 V standard NMOS thin-oxide devices with low threshold voltages. The reference current circuit includes transistors M_1 – M_{14} and M_{19} – M_{20} . In this circuit, thin-oxide NMOS transistor M_{13} and thick-oxide NMOS transistor M_{14} form a two-transistor SDMT structure to generate a CTAT voltage. Four identical native transistors, M_9 – M_{12} , with narrow channel widths and long channel lengths, are biased in the deep-linear region, effectively acting as large resistances. Proper transistor sizing allows the generation of a reference current I_{REF} with a near-zero temperature coefficient.

$$I_{\text{REF}} = \frac{1}{4} \mu_9 C_{\text{OX}9} K_9 m V_T \ln \left(\frac{\mu_7 C_{\text{OX}7} K_7}{\mu_6 C_{\text{OX}8} K_8} \right) \cdot \left[(V_{\text{th}16} - V_{\text{th}15} - V_{\text{th}9}) + m V_T \ln \left(\frac{\mu_5 C_{\text{OX}15} K_{15}}{\mu_6 C_{\text{OX}16} K_{16}} \right) \right] \quad (1)$$

Where, μ denotes the carrier mobility, C_{OX} is the gate-oxide capacitance per unit area, W/L represents the MOSFET width-to-length ratio, K is the transistor size scaling factor, m

This work was supported by Shenzhen Science and Technology Program under Grant JCYJ20250604175601003 and ZDCY20250901104259009.

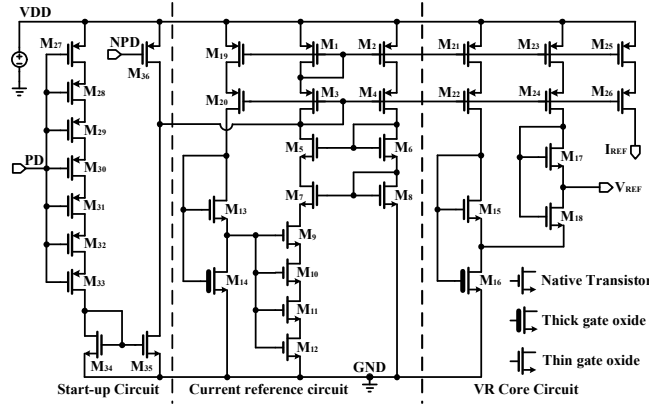


Fig.1. The circuit diagram of the proposed subthreshold CMOS-only voltage reference.

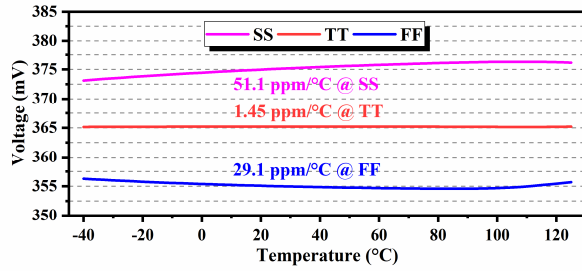


Fig.2. The simulation results of the proposed VR output voltage V_{REF} over process corners.

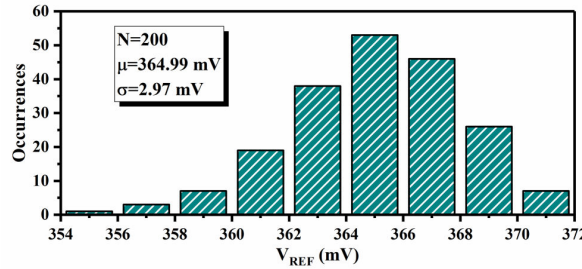


Fig.3. The Monte Carlo simulation results of the proposed VR output voltage V_{REF} .

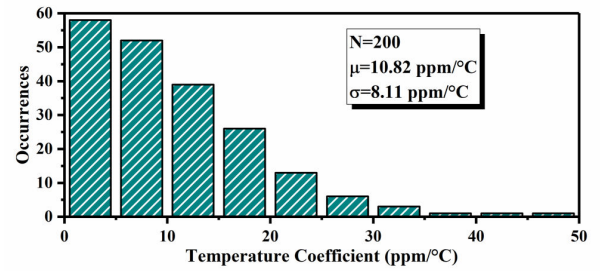


Fig.4. The Monte Carlo simulation results of TC for the proposed VR output voltage V_{REF} .

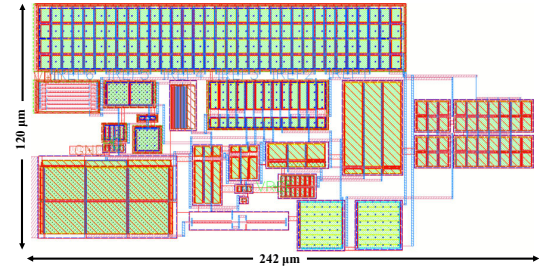


Fig.5. The layout of the proposed VR circuit.

is the subthreshold slope factor, V_T is the thermal voltage, and V_{th} is the threshold voltage.

The reference voltage V_{REF} core circuit is implemented using a two-stage SDMT structure. In the first stage, the SDMT comprises a 3.3 V standard NMOS thick-oxide transistor M_{16} with a high threshold voltage and a 1.8 V standard NMOS thin-oxide transistor M_{15} with a low threshold voltage. The second-stage SDMT consists of two 1.8 V standard NMOS thin-oxide transistors, M_{17} and M_{18} , both with low threshold voltages. A cascode current mirror formed by transistors M_{21} – M_{24} replicates the zero-TC reference current to provide biasing for the VR core. By appropriately setting the current mirroring ratio, the current through M_{15} is made equal to that through M_{17} and M_{18} , resulting in a current through M_{16} that is twice that of M_{17} and M_{18} . Assuming negligible body effect and identical subthreshold slope factors (m) for all transistors, the output voltage V_{REF} can be expressed using the MOSFET subthreshold current model as follows:

$$V_{REF} = (V_{th16} - V_{th15}) + mV_T \ln \left(\frac{2\mu_{A5}\mu_{17}C_{OX15}C_{OX17}K_{15}K_{17}}{\mu_{A6}\mu_{18}C_{OX16}C_{OX18}K_{16}K_{18}} \right) \quad (2)$$

By appropriately sizing the MOSFETs according to (2), a temperature-independent output voltage V_{REF} can be achieved. In practical applications, however, small variations in the subthreshold slope factor m among MOSFETs can affect the stability of branch bias currents, which in turn impacts the stability of the reference output voltage. In the proposed VR circuit, these bias currents are supplied by a stable reference current circuit, improving overall output stability and temperature performance while extending the operational temperature range. The first-stage SDMT generates both CTAT and PTAT voltages. In the second-stage SDMT, the threshold voltages of transistors of the same type cancel each other, producing only a PTAT voltage that compensates for temperature deviations introduced by the first stage. By combining coarse adjustment in the first stage with fine adjustment in the second stage, the two-stage SDMT structure achieves a stable and temperature-independent reference output voltage V_{REF} .

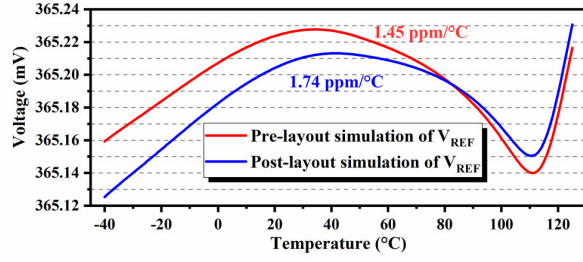


Fig. 6. Post-layout simulation results of the proposed VR output voltage V_{REF} .

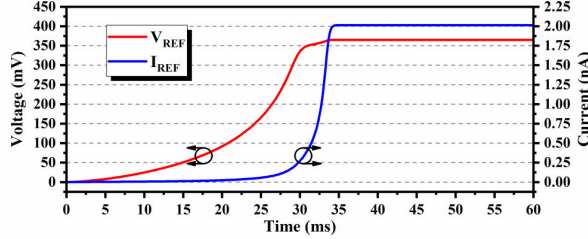


Fig. 7. Post-layout simulation results of the start-up circuit.

III. SIMULATION RESULTS OF THE PROPOSED VOLTAGE REFERENCE CIRCUIT

Fig. 2 shows the process corner simulation results of the proposed subthreshold CMOS-only VR circuit. The output voltage is 365 mV under the TT process corner at room temperature. Over a wide temperature range from -45°C to 125°C , the circuit achieves a TC of $1.45\text{ ppm}/^{\circ}\text{C}$. Under the FF corner, the TC increases to $29.1\text{ ppm}/^{\circ}\text{C}$ across the same temperature range, and further rises to $51.1\text{ ppm}/^{\circ}\text{C}$ under the SS corner.

To emulate the random variations inherent in semiconductor fabrication, Monte Carlo pre-simulations were conducted on the proposed VR circuit under a 1 V supply and room temperature conditions. As shown in Fig. 3, 200 Monte Carlo iterations yielded a mean reference output voltage of 364.99 mV with a standard deviation of 2.97 mV, corresponding to a σ/μ ratio of 0.81%. Fig. 4 illustrates the TC distribution obtained from the same 200 iterations, showing a mean TC of $10.82\text{ ppm}/^{\circ}\text{C}$ and a standard deviation of $8.11\text{ ppm}/^{\circ}\text{C}$.

The proposed VR circuit is all MOSFETs and designed in a standard $0.18\text{ }\mu\text{m}$ CMOS process. Fig. 5 shows the circuit layout, which occupies an area of 0.029 mm^2 . Fig. 6 presents the post-layout simulation results of the output voltage V_{REF} , demonstrating a TC of $1.74\text{ ppm}/^{\circ}\text{C}$ over a wide temperature range from -45°C to 125°C . The voltage reference operates as a circuit without external excitation. When power voltage is first applied, the circuit exhibits a metastable operating point corresponding to a zero-current state. Therefore, a start-up circuit is required to overcome this condition and allow the reference circuit to reach a stable operating state. In this paper, the start-up circuit consists of transistors $M_{27}\text{--}M_{36}$, as shown in Fig. 2. Transistors $M_{27}\text{--}M_{33}$ are ratioed devices with very small width-to-length ratios, generating minimal branch current to reduce overall power consumption. The enable terminal PD is active low during normal operation and disables the circuit when driven

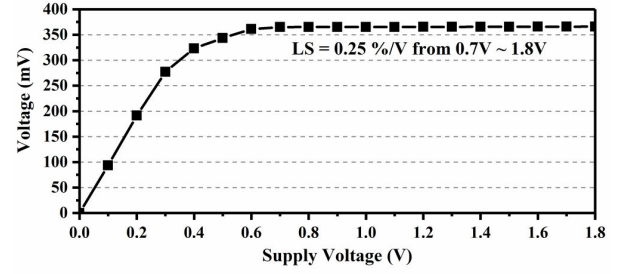


Fig. 8. Post-layout simulation results of the LS of the proposed VR.

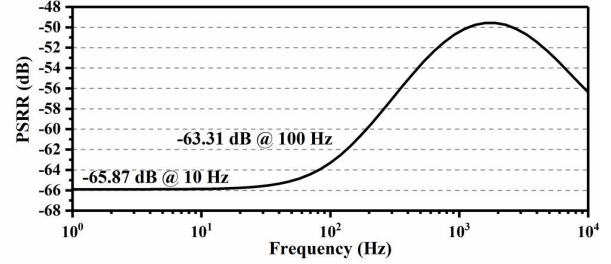


Fig. 9. Post-layout simulation results of the PSRR of the proposed VR.

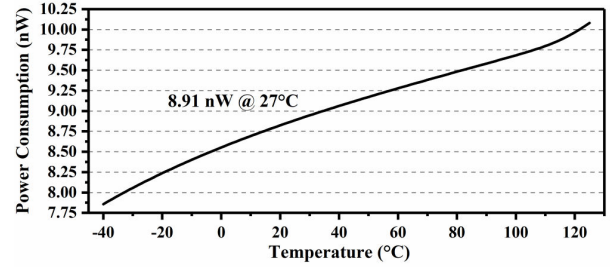


Fig. 10. Post-layout simulation results of the power consumption.

high. During power-up, transistors $M_{27}\text{--}M_{33}$ turn on, allowing current to flow into transistor M_{34} . This current is mirrored by M_{35} , while the NPD signal remains high, keeping M_{36} turned off. As a result, the gate voltages of M_3 and M_4 begin to rise, initiating a nonzero current in the reference loop. The current reference formed by transistors $M_1\text{--}M_{20}$ gradually becomes active, enabling the circuit to reach its steady-state operating condition. Fig. 7 shows the post-layout simulation results of the start-up circuit, confirming that both the current and voltage reference circuits can start reliably and achieve normal operation.

The post-layout simulation results of the proposed VR circuit's LS at room temperature are shown in Fig. 8, illustrating how the reference output varies with changes in the DC supply voltage. When the supply voltage is below 0.7 V, the output voltage increases proportionally. Above 0.7 V, the reference voltage stabilizes. Within a supply range of 0.7–1.8 V, the VR circuit achieves an LS of 0.25 %/V . Fig. 9 presents the simulated PSRR of the circuit. The PSRR is 65.87 dB at 10 Hz and gradually decreases with increasing frequency, reaching 63.31 dB at 100 Hz. The temperature dependence of the circuit's power consumption under a 1 V supply is shown in Fig. 10. Post-layout simulations indicate a total current consumption of 8.91 nA at room temperature,

TABLE I. PERFORMANCE OF THIS WORK COMPARED WITH OTHER REFERENCES

	This work ^a	[5] ^a	[7] ^a	[11] ^b	[13] ^a	[14] ^b	[15] ^a
Tech. (nm)	180	130	180	180	180	180	180
Type	CMOS	Sub-BGR	CMOS	CMOS	CMOS	CMOS	CMOS
Min VDD (V)	0.8	0.9	0.7	0.9	1.0	0.25	0.8
V_{REF} (mV)	365	474	147	261	346	118.1	332.85
LS (%/V)	0.25	0.10	0.031	0.013	0.23	0.3	0.5
PSRR (dB)@100Hz	63.31	NA	NA	73.5	63	65	31
Temp. Range (°C)	-40 ~ 125	-20 ~ 80°C	-40 ~ 125	-40 ~ 130	-40 ~ 140	-40 ~ 140	0 ~ 120
Area (mm ²)	0.029	0.040	0.01	0.005	0.45	0.001	NA
Power (nW)	8.91	30	21	1.8	200	0.113	15.6
TC	1.74	19.0	66.38	62	9.8	73.5	77.425

^a Simulation results. ^b Measured results.

corresponding to a power dissipation of 8.91 nW. Table I summarizes the performance comparison between the proposed design and state-of-the-art voltage reference circuits. The proposed VR outperforms existing designs in terms of power consumption and temperature coefficient, while also achieving excellent PSRR and occupying a smaller silicon area. Overall, the proposed VR circuit demonstrates superior performance, providing an effective solution for low-power, high-precision temperature sensing applications.

IV. CONCLUSION

This paper presents a subthreshold CMOS-only voltage reference circuit designed using a standard 0.18 μm CMOS process, occupying a compact active area of 0.029 mm². The circuit delivers a stable reference voltage of 365 mV and achieves a temperature coefficient of 1.74 ppm/°C across a wide temperature range from -40 °C to 125 °C, demonstrating excellent thermal stability. Post-layout simulations indicate that the proposed design attains a PSRR of 63.31 dB at 100 Hz and a line sensitivity of 0.25 %/V, ensuring strong immunity to supply and process variations. Owing to its all-MOS architecture and subthreshold operation, the circuit exhibits ultra-low power consumption while maintaining high precision, making it highly suitable for integration in low-power and high-reliability temperature sensing or environmental monitoring systems operating under harsh or resource-limited conditions.

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